Question

Element $\mathbf{X}$ can react with AgF under strictly dry and low-pressure conditions to obtain an isomer $\mathbf{A}_1$ of compound $\mathbf{A}$ with C_2 symmetry. Another isomer $\mathbf{A}_2$ can be obtained by reacting $\mathrm{KSO_3F}$ with $\mathbf{B}$ in liquid $\mathrm{SO_2}$. $\mathbf{A}_2$ is thermodynamically unstable and can disproportionate into compound $\mathbf{C}$ and elemental $\mathbf{X}$ under HF catalysis. $\mathbf{C}$ can form a 1:1 adduct $\mathbf{D}$ with CsF. Chlorination of $\mathbf{B}$ in the presence of $\mathrm{FeCl_3}$ yields compound $\mathbf{E}$. Reacting $\mathbf{E}$ with NaF in warm diethyl ether yields a mixture of $\mathbf{B}$ and $\mathbf{C}$. Both $\mathbf{B}$ and $\mathbf{E}$ can react with $\mathrm{NH_4Cl}$ to produce a cyclic compound $\mathbf{F}$. $\mathbf{F}$ is insoluble in water but soluble in concentrated NaOH solution to form two sodium salts $\mathbf{G}$ and $\mathbf{H}$. The mass fractions of Na in $\mathbf{G}$ and $\mathbf{H}$ are 29.09 % and 36.48 %, respectively. It is known that all substances represented by letters in the question contain $\mathbf{X}$. From the following options, choose the one that best fits the meaning of the question.

A. All other options are incorrect

B. $\mathbf{A}_1$ and $\mathbf{A}_2$ have different structures, but both possess a diad axis.

C. $\mathbf{A}_2$ and $\mathbf{B}$ have different structures, but their symmetry elements are completely identical.

D. The structure of $\mathbf{C}$ and the corresponding structure of the substance in $\mathbf{D}$ that forms a 1:1 adduct, although slightly different, share the same hybridization mode of the central atom.

E. The structure of $\mathbf{C}$ and the corresponding structure of the substance in $\mathbf{D}$ that forms a 1:1 adduct, although slightly different in the hybridization mode of the central atom, share the same symmetry elements.

F. The valence states of $\mathbf{C}, \mathbf{E}$ are consistent with those of the cyclic compound $\mathbf{F}$ formed.

G. The valence states of $\mathbf{C}$ and $\mathbf{E}$, as well as the valence state of the formed cyclic compound $\mathbf{F}$, are all distinct from each other.

Answer

Correct Answer: G

Detailed Explanation

The breakthrough point of the problem lies in the information of compounds $\mathbf{G}$ and $\mathbf{H}$.

Calculating the molar mass of $\mathbf{G}$ and $\mathbf{H}$:

For $\mathbf{G}$: The mass fraction of sodium is 29.09%.

The molar mass of G satisfies:

If $n = 1$, $M(\mathbf{G}) \approx 79.03\mathrm{g/mol}$, the mass of the anion part $\approx 79.03 - 22.99 = 56.04\mathrm{g/mol}$.

If $n = 2$, $M(\mathbf{G}) \approx 158.06\mathrm{g/mol}$, the mass of the anion part $\approx 158.06 - 45.98 = 112.08\mathrm{g/mol}$.

If $n = 3$, $M(\mathbf{G}) \approx 237.09\mathrm{g/mol}$, the mass of the anion part $\approx 237.09 - 68.97 = 168.12\mathrm{g/mol}$.

Because it is treated in concentrated NaOH solution, and the anion is composed of elements $\mathbf{X}$ and O, $\mathbf{G}$ is $\mathrm{Na_2S_2O_3}$.

CHECKPOINT 1 PTS

$\mathbf{G}$ is $\mathrm{Na_2S_2O_3}$

Similarly, for $\mathbf{H}$: The mass fraction of sodium is 36.48%, calculated to be $\mathbf{H}$ is $\mathrm{Na_2SO_3}$.

CHECKPOINT 1 PTS

$\mathbf{H}$ is Na_2SO_3

Therefore, element $\mathbf{X}$ is S.

$\mathbf{F}$ is hydrolyzed in $NaOH$ to obtain Na_2SO_3 and $Na_2S_2O_3$. We know that $\mathbf{F}$ is a cyclic compound, so we can deduce that $\mathbf{F}$ is cyclic S_4N_4.

CHECKPOINT 1 PTS

$\mathbf{F}$ is cyclic S_4N_4

$\mathbf{B}$ or $\mathbf{E}$ can react with NH_4Cl to obtain S_4N_4; $\mathbf{B}$ is chlorinated in the presence of $FeCl_3$ to obtain $\mathbf{E}$, one of which is S_2Cl_2, and the other should be SCl_2. The latter has a higher valence, so $\mathbf{B} = S_2Cl_2$, $\mathbf{E} = SCl_2$.

CHECKPOINT 1 PTS

$\mathbf{B} = S_2Cl_2$

CHECKPOINT 1 PTS

$\mathbf{E} = SCl_2$

Elemental sulfur reacts with AgF to obtain $\mathbf{A}$, whose isomers $\mathbf{A}_1$ belong to the C_2 point group and $\mathbf{A}_2$ belong to the C_s point group, that is, $\mathbf{A}_1 = \text{F-S-S-F}$, $\mathbf{A}_2 = \mathrm{S{=}SF_2}$. It can be known that the symmetry elements of $\mathbf{A}_2$ and $\mathbf{A}_1, \mathbf{B}$ are different.

CHECKPOINT 1 PTS

$\mathbf{A}$ is $\mathrm{S_2F_2}$

CHECKPOINT 1 PTS

$\mathbf{A}_1$ is F-S-S-F, belonging to the C_2 point group

CHECKPOINT 1 PTS

$\mathbf{A}_2 = \mathrm{S{=}SF_2}$, belonging to the C_s point group

CHECKPOINT 1 PTS

The symmetry elements of $\mathbf{A}_2$ and $\mathbf{A}_1, \mathbf{B}$ are different

$\mathbf{A}_2$ disproportionates to produce $\mathbf{C}$ and S, so $\mathbf{C} = \mathrm{SF_4}$, and the hybridization is sp^3d hybridization.

CHECKPOINT 1 PTS

$\mathbf{C} = \mathrm{SF_4}$

According to the above information, we can know that the valences of $\mathbf{C}$, $\mathbf{E}$ and the valence of the cyclic compound $\mathbf{F}$ formed are all different from each other.

CHECKPOINT 1 PTS

The valences of $\mathbf{C}$, $\mathbf{E}$ and the valence of the cyclic compound $\mathbf{F}$ formed are all different from each other.

$\mathbf{C}$ reacts with CsF to form a $1:1$ adduct $\mathbf{D}$, that is, $\mathbf{D} = \mathrm{CsSF_5}$, and the hybridization is sp^3d^2 hybridization.

CHECKPOINT 1 PTS

$\mathbf{D} = \mathrm{CsSF_5}$

CHECKPOINT 1 PTS

The hybridization modes of S atoms in $\mathbf{C}$ and $\mathbf{D}$ are different, which are sp^3d hybridization and sp^3d^2 hybridization, respectively

CHECKPOINT 1 PTS

The anion symmetry of **C** and **D** are different